TASK 02 – IPv4 Addressing and Subnetting

TechBiz – CCNA Networking Internship
Week 1 | Task 2

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1. Introduction

This task focuses on IPv4 addressing, binary conversion, subnet masks and Variable Length Subnet Masking (VLSM). The practical component involved redesigning the network topology from Task 1 using five subnets allocated from the 192.168.10.0/24 network.

The network was implemented in Cisco Packet Tracer using three routers, three switches and four PCs. Static routing was configured to enable communication between all three LANs.

2. Objectives

- Practise converting decimal numbers into binary and binary numbers into decimal.
- Understand network addresses, broadcast addresses, usable host ranges and subnet masks.
- Divide a /24 network into five VLSM subnets based on host requirements.
- Re-address the existing Packet Tracer topology using the new subnet allocations.
- Configure routers, switches and PCs and verify end-to-end connectivity.

3. Binary Conversion

IPv4 addresses consist of 32 bits divided into four 8-bit octets. Each bit represents a power of two.

The place values of an 8-bit octet are:

128 | 64 | 32 | 16 | 8 | 4 | 2 | 1

A bit set to 1 contributes its corresponding value to the decimal number, while a bit set to 0 contributes nothing.

3.1 Decimal to Binary and Binary to Decimal

The following examples demonstrate both conversion directions.

Decimal	Working (sum of powers of 2)	8-bit binary
10	8 + 2	00001010
25	16 + 8 + 1	00011001
64	64	01000000
96	64 + 32	01100000
113	64 + 32 + 16 + 1	01110001
114	64 + 32 + 16 + 2	01110010
117	64 + 32 + 16 + 4 + 1	01110101
118	64 + 32 + 16 + 4 + 2	01110110
192	128 + 64	11000000
255	128 + 64 + 32 + 16 + 8 + 4 + 2 + 1	11111111

3.2 Binary to Decimal Verification

To convert binary back to decimal, the values of the positions containing 1 are added together.

Binary	Calculation	Decimal
00001010	8 + 2	10
00011001	16 + 8 + 1	25
01000000	64	64
01100000	64 + 32	96
01110001	64 + 32 + 16 + 1	113
01110010	64 + 32 + 16 + 2	114
01110101	64 + 32 + 16 + 4 + 1	117
01110110	64 + 32 + 16 + 4 + 2	118
11000000	128 + 64	192
11111111	128 + 64 + 32 + 16 + 8 + 4 + 2 + 1	255

These conversions demonstrate how binary values represent decimal IPv4 octets.

4. IPv4 Addressing Concepts

Network Address: Identifies a subnet. It is the first address in the subnet and cannot normally be assigned to a host.

Broadcast Address: The last address in an IPv4 subnet. It is used to send traffic to all hosts within that subnet.

Usable Host Range: The addresses between the network and broadcast addresses that can be assigned to devices.

Subnet Mask: Identifies the network and host portions of an IPv4 address. For example, 255.255.255.192 represents a /26 prefix, meaning 26 network bits and 6 host bits.

5. VLSM Subnetting Plan

The original network was 192.168.10.0/24.

Five subnets were required for the following host requirements:

- 50 hosts
- 25 hosts
- 12 hosts
- 2 hosts
- 2 hosts

VLSM allocates subnet sizes according to the number of hosts required. The largest subnet is allocated first to use address space efficiently.

5.1 VLSM Calculations

Host requirement	Prefix	Total addresses	Usable hosts
50	/26	64	62
25	/27	32	30
12	/28	16	14
2	/30	4	2
2	/30	4	2

The allocation follows the required block sizes:

- /26: 256 − 192 = 64 addresses.
- /27: 256 − 224 = 32 addresses.
- /28: 256 − 240 = 16 addresses.
- /30: 256 − 252 = 4 addresses.

5.2 Final VLSM Allocation

Requirement	Network address	Subnet mask	Usable host range	Broadcast
50 hosts	192.168.10.0/26	255.255.255.192	.1 – .62	.63
25 hosts	192.168.10.64/27	255.255.255.224	.65 – .94	.95
12 hosts	192.168.10.96/28	255.255.255.240	.97 – .110	.111
2 hosts	192.168.10.112/30	255.255.255.252	.113 – .114	.115
2 hosts	192.168.10.116/30	255.255.255.252	.117 – .118	.119

The subnets are consecutive, correctly aligned and non-overlapping. All five fit within the original /24 network.

6. Practical Implementation – Cisco Packet Tracer

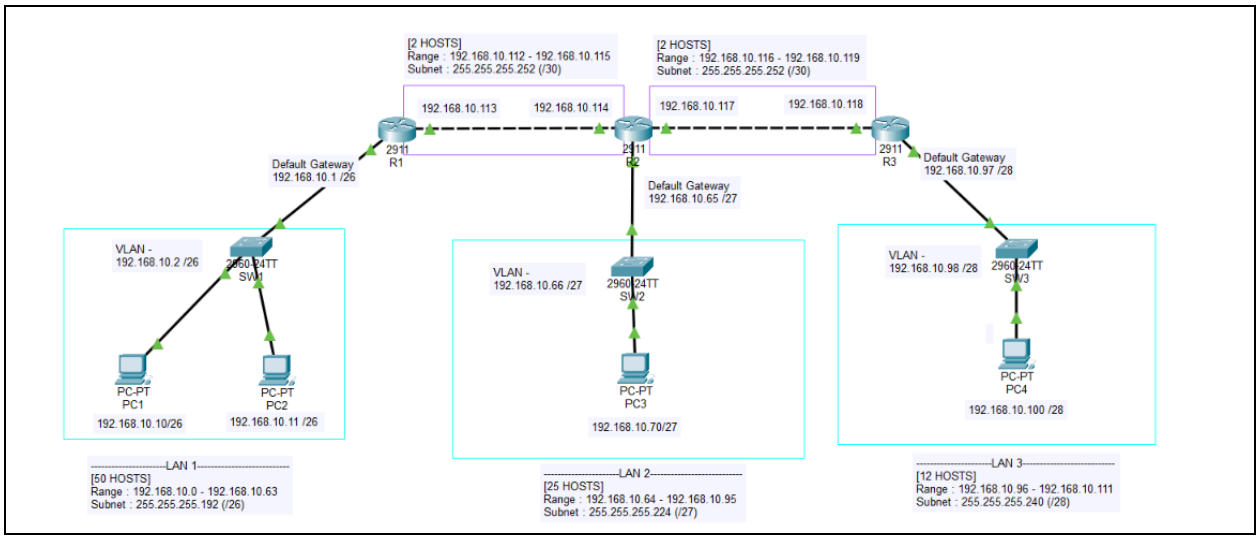
6.1 Network Topology

The original Task 1 topology was expanded to include a third router and a third switch.

Devices used:

- 3 × Cisco 2911 Routers
- 3 × Cisco 2960 Switches
- 4 × PCs

The topology consists of three separate LANs connected through three routers. The two /30 subnets are used for point-to-point router connections.



6.2 Device IP Addressing

The following addresses were configured on the routers:

Router	Interface	IP address	Subnet
R1	G0/0	192.168.10.1	/26
R1	G0/1	192.168.10.113	/30
R2	G0/0	192.168.10.65	/27
R2	G0/1	192.168.10.114	/30
R2	G0/2	192.168.10.117	/30
R3	G0/0	192.168.10.97	/28
R3	G0/1	192.168.10.118	/30

The switch management addresses were configured as follows:

Switch	VLAN 1 IP address	Subnet
SW1	192.168.10.2	/26
SW2	192.168.10.66	/27
SW3	192.168.10.98	/28

The PCs were assigned the following addresses:

PC	IP address	Subnet mask	Default gateway
PC1	192.168.10.10	255.255.255.192	192.168.10.1
PC2	192.168.10.11	255.255.255.192	192.168.10.1
PC3	192.168.10.70	255.255.255.224	192.168.10.65
PC4	192.168.10.100	255.255.255.240	192.168.10.97

6.3 Router Configuration and Static Routing

The router interfaces were configured using Cisco IOS CLI commands, including:

- interface to select the required interface.
- ip address to assign an IPv4 address and subnet mask.
- no shutdown to enable the interface.
- show ip interface brief to verify interface status.

Static routes were configured to enable communication between the three LANs.

R1:

- Route to 192.168.10.64/27 via 192.168.10.114.
- Route to 192.168.10.96/28 via 192.168.10.114.

R2:

- Route to 192.168.10.0/26 via 192.168.10.113.
- Route to 192.168.10.96/28 via 192.168.10.118.

R3:

- Route to 192.168.10.0/26 via 192.168.10.117.
- Route to 192.168.10.64/27 via 192.168.10.117.

The show ip route command was used to verify directly connected and static routes.

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.10.113  YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 6 subnets, 5 masks
C       192.168.10.0/26 is directly connected, GigabitEthernet0/0
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0
S       192.168.10.64/27 [1/0] via 192.168.10.114
S       192.168.10.96/28 [1/0] via 192.168.10.114
C       192.168.10.112/30 is directly connected, GigabitEthernet0/1
L       192.168.10.113/32 is directly connected, GigabitEthernet0/1
```

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 8 subnets, 5 masks
S       192.168.10.0/26 [1/0] via 192.168.10.113
C       192.168.10.64/27 is directly connected, GigabitEthernet0/0
L       192.168.10.65/32 is directly connected, GigabitEthernet0/0
S       192.168.10.96/28 [1/0] via 192.168.10.118
C       192.168.10.112/30 is directly connected, GigabitEthernet0/1
L       192.168.10.114/32 is directly connected, GigabitEthernet0/1
C       192.168.10.116/30 is directly connected, GigabitEthernet0/2
L       192.168.10.117/32 is directly connected, GigabitEthernet0/2
```

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.10.0/24 is variably subnetted, 6 subnets, 5 masks
S       192.168.10.0/26 [1/0] via 192.168.10.117
S       192.168.10.64/27 [1/0] via 192.168.10.117
C       192.168.10.96/28 is directly connected, GigabitEthernet0/0
L       192.168.10.97/32 is directly connected, GigabitEthernet0/0
C       192.168.10.116/30 is directly connected, GigabitEthernet0/1
L       192.168.10.118/32 is directly connected, GigabitEthernet0/1
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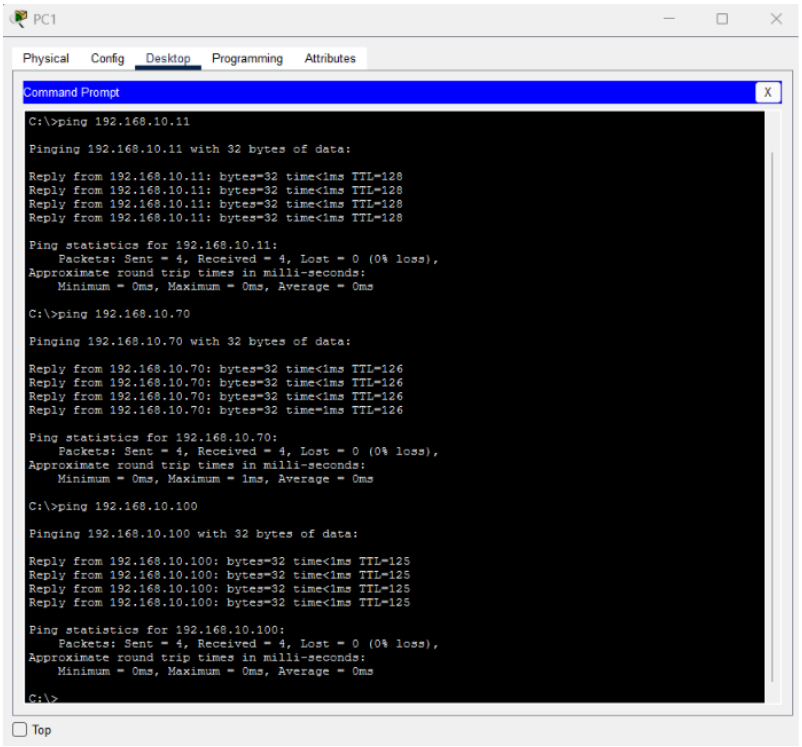
7. Connectivity Testing

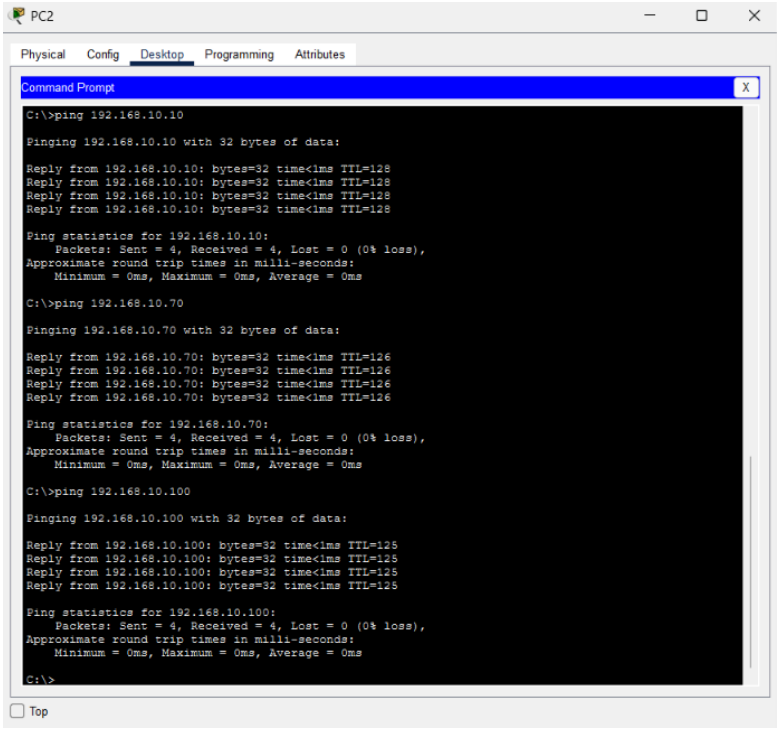
Connectivity was verified using the ping command from the command prompt of each PC.

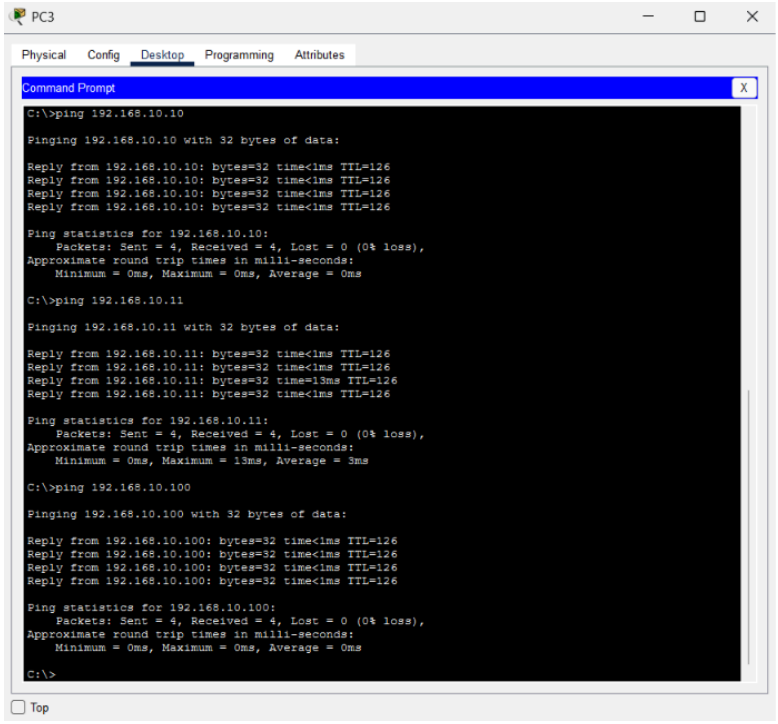
The tests included:

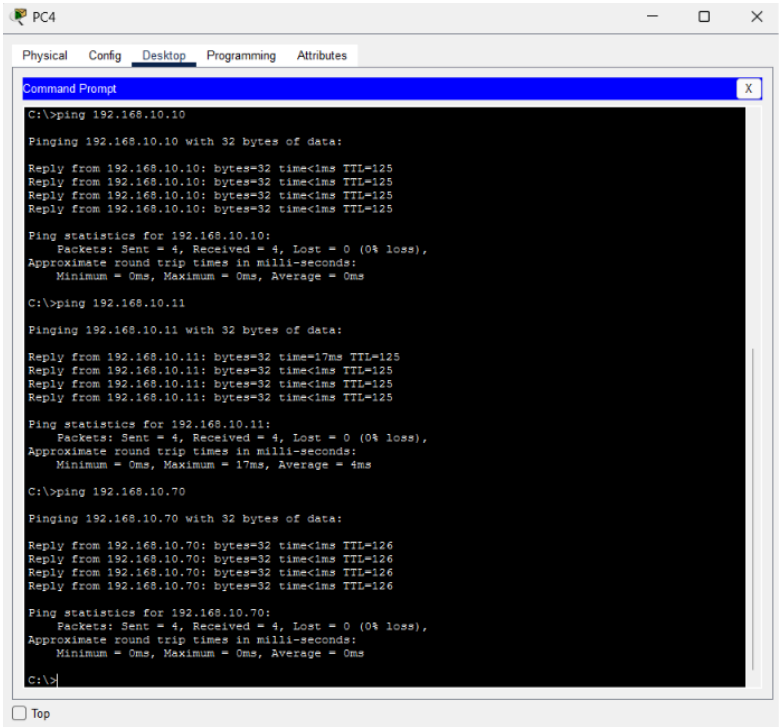
- Communication between PCs within the same LAN.
- Communication between PCs in different LANs.
- Communication across the three-router topology.

All four PCs successfully communicated with the other PCs. The successful ICMP ping responses confirmed that the IP addressing, default gateways and static routing configurations were working correctly.









8. Conclusion

This task provided practical experience with IPv4 addressing, binary conversion and VLSM subnetting. The original /24 network was successfully divided into five subnets of different sizes according to their host requirements.

The redesigned Packet Tracer lab demonstrated the use of separate LANs, point-to-point router links, subnet masks, default gateways and static routing.

Connectivity tests confirmed successful communication between all four PCs across the three LANs. The task strengthened my understanding of subnet allocation, address planning and routing between different IPv4 networks.

Tools Used: Cisco Packet Tracer, Cisco IOS CLI
Internship: TechBiz – CCNA Networking
Task: 02 – IPv4 Addressing, Binary & VLSM Subnetting